

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI

AIM : Adaboost Ensemble Learning

Implement the Adaboost algorithm to create an ensemble of weak classifiers.

Train the ensemble model on a given dataset and evaluate its performance.

Compare the results with individual weak classifiers.

CODE:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Load dataset
data = pd.read_csv("telescope_data.csv")

# Remove unnecessary index column
data = data.drop(columns=["Unnamed: 0"], errors="ignore")

# Separate features and target
X = data.drop(columns=["class"])
y = data["class"].map({"g": 1, "h": 0})

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# -----
# Individual Weak Classifier: Decision Stump
# -----
weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_classifier.fit(X_train, y_train)

# Prediction using weak classifier
```

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```
y_pred_weak = weak_classifier.predict(X_test)

# Evaluate weak classifier
weak_accuracy = accuracy_score(y_test, y_pred_weak)

print("=" * 60)
print("INDIVIDUAL WEAK CLASSIFIER - DECISION STUMP")
print("=" * 60)
print("Accuracy:", round(weak_accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_weak))
print("\nClassification Report:")
print(classification_report(y_test, y_pred_weak))

# -----
# AdaBoost Ensemble
# -----
adaboost = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1, random_state=42),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

adaboost.fit(X_train, y_train)

# Prediction using AdaBoost
y_pred_ada = adaboost.predict(X_test)

# Evaluate AdaBoost
ada_accuracy = accuracy_score(y_test, y_pred_ada)

print("=" * 60)
print("ADABOOST ENSEMBLE CLASSIFIER")
print("=" * 60)
print("Accuracy:", round(ada_accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_ada))
print("\nClassification Report:")
```

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```
print(classification_report(y_test, y_pred_ada))

# -----
# Comparison
# -----

print("=" * 60)
print("PERFORMANCE COMPARISON")
print("=" * 60)
print(f"Decision Stump Accuracy : {weak_accuracy * 100:.2f}%")
print(f"AdaBoost Accuracy      : {ada_accuracy * 100:.2f}%")
print(f"Improvement             : {(ada_accuracy - weak_accuracy) * 100:.2f}%")

# -----
# Accuracy Comparison Graph - ADDED
# -----

import matplotlib.pyplot as plt

models = ["Decision Stump", "AdaBoost"]
accuracies = [weak_accuracy * 100, ada_accuracy * 100]

plt.figure(figsize=(7, 5))
bars = plt.bar(models, accuracies)

plt.title("Decision Stump vs AdaBoost Accuracy")
plt.xlabel("Classifier")
plt.ylabel("Accuracy (%)")
plt.ylim(0, 100)

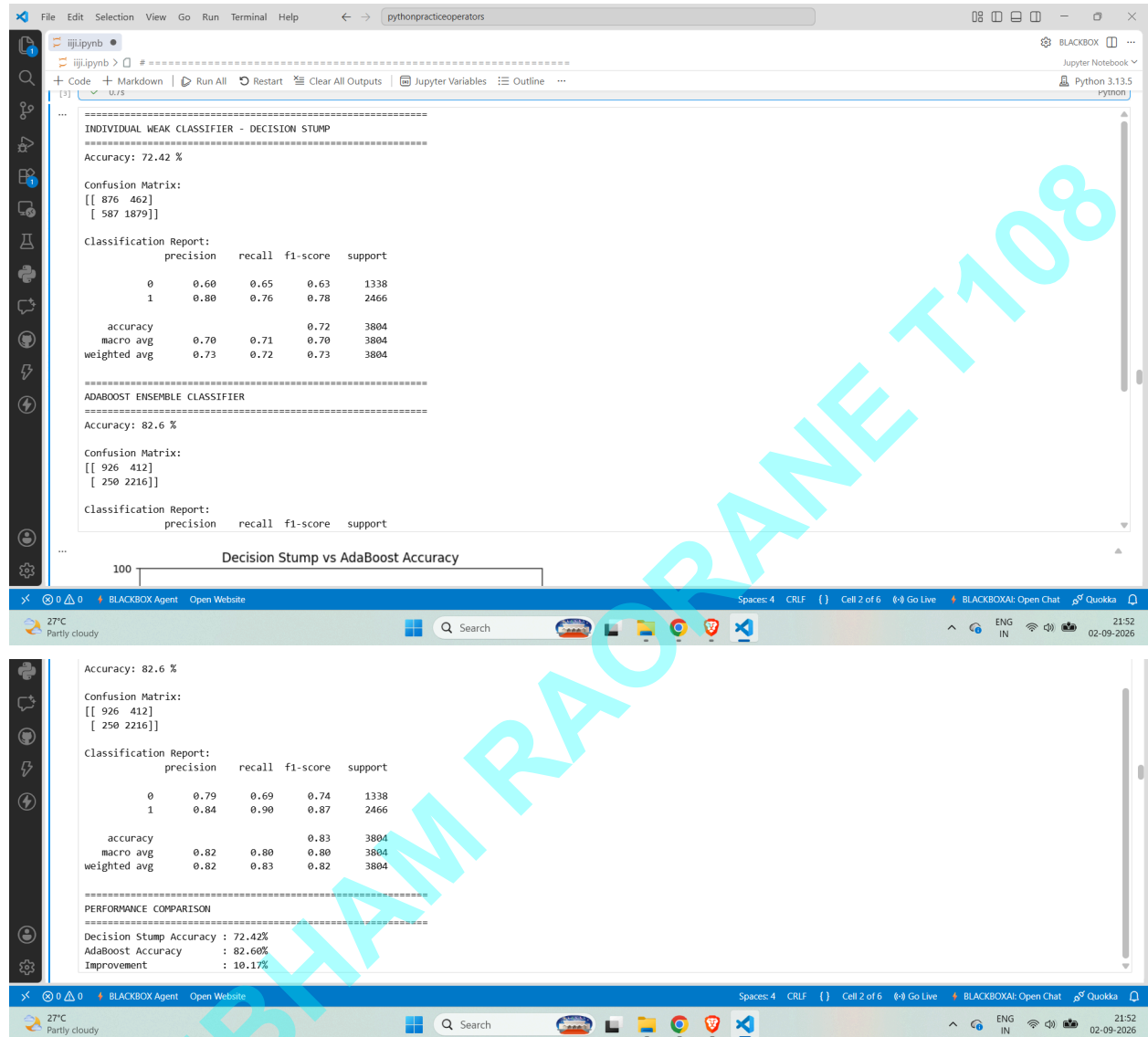
for bar, accuracy in zip(bars, accuracies):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 1,
        f"{accuracy:.2f}%",
        ha="center"
    )

plt.show()
```

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OUTPUT



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